

OSCE

Nutrition

SAM

- Admission criteria
 - Depends on age
 - Weight for length
 - Edema and
 - Complications presence or absence



- Picture with wasted children
- What do you observe?
- How do you approach this patient?
- What investigation do you order?
- What are the complications?
- Write the outline of management?

- **Marasmus**
- GA-irritability, Wasted, ematiated,
- HEENT-little old man or monkey face, no buccal fat, zygomatic prominence, NO SKIN CHANGE BUT MILD HAIR CHANGE, vit A deficiency signs, anemia, dry buccal mucosa, pale conjunctiva,
- angular chelosis(ddx- Fe def anemia, riboflavin def, oral candidiasis), tongue smoothness(due to Fe def), papillary atrophy
- Anthropometry- wt for age less than 60 and no edema, MUAC les than 11cm
- Protuberant abdomen- pot belly abdomen
- Baggy pant appearance
- Hypoglycemia, hypothermia

- Kwashiorkor
- GA-moon face, apathetic, irritable
- Hair-sparse, thin, loss of luster, easily pluckable, grayish in color, Flag sign(black, grey, red hair color)
- Presence of SC fat, but there is wasting.
- Puffy face with periorbital edema
- Skin- flaky paint dermatitis, peeling dermatitis- ulcer
- Abd distention and hepatomegaly
- Edema
- Signs of anemia

Age 0- 6 month

Sign	Classify	Treat
1. WFL < -3 Z Score , and • Presence of complications Or • Edema of both feet	Complicated Severe Acute Malnutrition	• In-patient management (CARE PLAN C)
2. WFL < -3 Z score , and • No complications Or/And • No edema of both feet	Uncomplicated severe acute malnutrition	• Counsel on breast feeding and care. • Orphans other option of feeding
3. WFL $\geq -3 Z$ to $\leq -2 Z$ score And • No edema of both feet	Moderate Acute Malnutrition	
4. WFL $\geq -2 Z$ score And • No edema of both feet	No Acute Malnutrition	• Congratulate • Counsel on feeding (CARE PLAN A)

6 month to 59 month

Signs	Classify	Treat
1. WFL/H < -3Z score OR • MUAC <11.5 cm** OR • Edema of both feet (+, ++), PLUS a. Any one of the medical complications , or b. Failed Appetite test	Complicated Severe Acute Malnutrition	• Admit for in-patient management (CARE PLAN C-Inpatient)
2. +++ Edema, OR • WFL/H < -3Z with edema (Marasmic Kwashiorkor), OR • MUAC <11.5 cm with edema		• Manage in OTP OR • Manage as in-patient if OTP service is not available (CARE PLAN C outpatient)

Continuation for 6-59 month

Sign	Classify	Treat
3. WFL/H < -3Z score OR MUAC < 11.5 cm OR Edema of both feet (+, ++) AND No medical complication AND pass appetite test	Uncomplicated Severe Acute Malnutrition	Manage in OTP - using the OTP protocol or - manage as in-patient if OTP service is not available (CARE PLAN C-outpatient)

Continuation 2

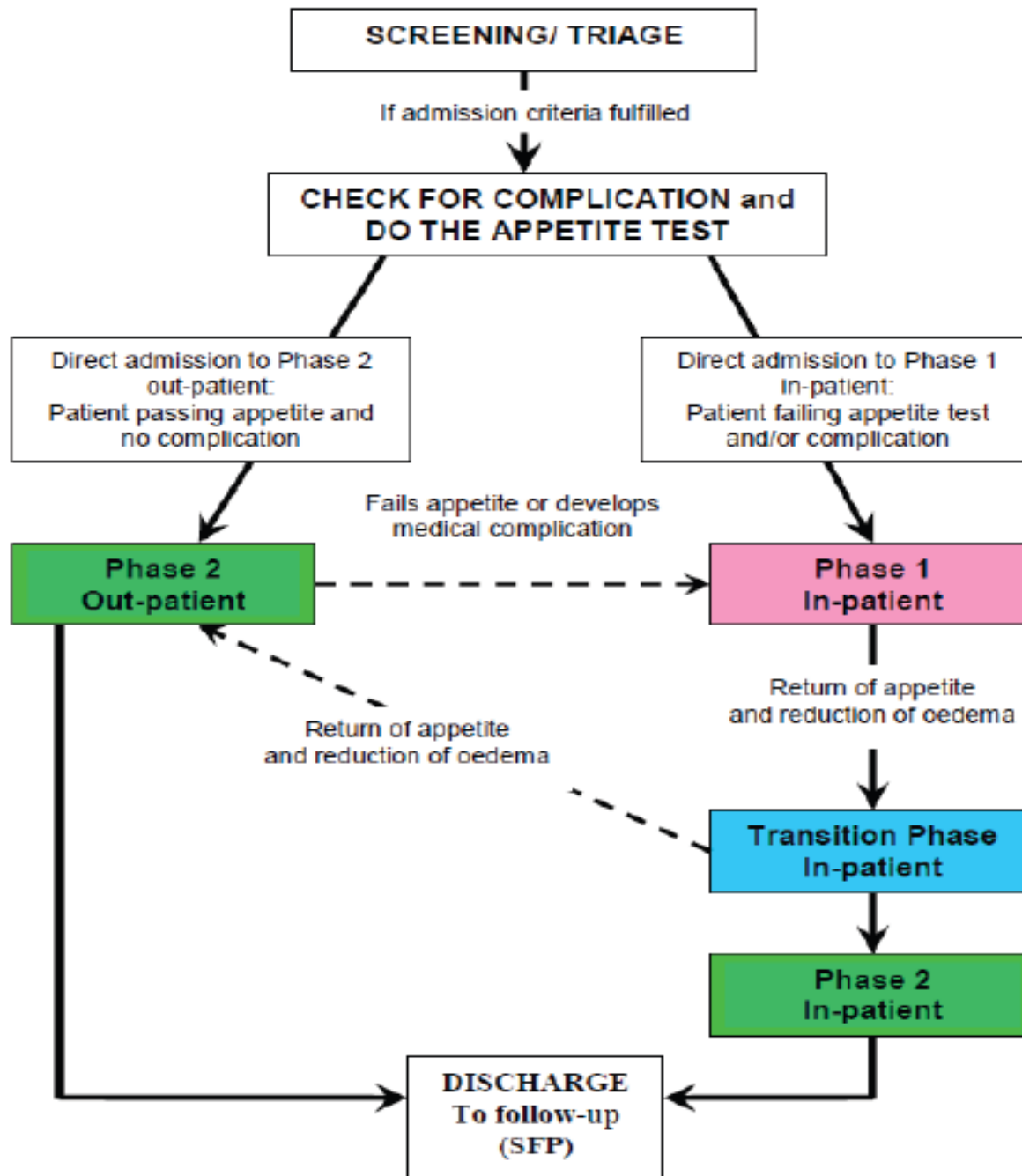
Signs	Classify	Treat
4. WFL/H $\geq$ -3Z to < -2Z score OR - MUAC 11.5 cm to <12.5 cm AND - No edema of both feet	Moderate Acute Malnutrition	- Admit to SFP, if available (Supplementary Feeding Program), - Counsel on infant and child feeding/care (CARE PLAN B)
5. If WFL/H $\geq$ -2Z score OR - MUAC $\geq$ 12.5 cm AND - No edema of both feet	No Acute Malnutrition	Congratulate and Counsel the mother on infant and child feeding/care (CARE PLAN A)

5-18 years old

1. BMI for age <-3SD OR ☐ Edema of both feet with no other cause (+, ++), PLUS • <i>Any one of the medical complications</i> , or • <i>Failed Appetite test</i> OR ☐ (+++) Edema,	Complicated Severe Acute Malnutrition	Admit for in-patient management (CARE PLAN C-Inpatient)
2. BMI for age <-3SD OR ☐ Edema of both feet with no other cause (+, ++) AND ☐ any other medical condition	Uncomplicated Severe Acute Malnutrition	Manage in OTP using the OTP protocol or • <i>manage as in-patient if OTP service is not available</i>

Continuation for 5-18 years

3. BMI for age -3 to <2 SD category OR AND No edema of both feet	Moderate Acute Malnutrition	Admit to SFP if available Counsel on infant and child feeding/care (CARE PLAN B)
4. BMI for age > -2 SD AND No edema of both feet	No Acute Malnutrition	Congratulate and Counsel the mother on infant and child feeding/care (CARE PLAN A)



Other Clinical features

Edematous SAM

(Kwashiorkor)

- Age: commonly 2-3yrs
- Apathy
- "Radar Gaze"
- "Moon Face"— sagging cheeks
- Edema
- Liver enlargement
- Dermatosis



Nonedematous SAM

(Marasmic)

- Age:- < 5yrs, but any age
- Irritable / apathic
- Sever growth retardation
- No edema
- Wrinkled skin & Bony prominence



Apetitie test

- Use plumpy nut and check how much of the feed they finish

The Minimum Amount	
<i>Plumpy'nut</i>	
Body weight (Kg)	Sachets
Less than 4 kg	1/8 to 1/4
4 – 6.9	1/4 to 1/3
7 – 9.9	1/3 to 1/2
10 – 14.9	1/2 to 3/4
15 - 29	3/4 to 1
Over 30 kg	>1

It is mainly metabolic malnutrition that causes death.

A poor appetite means that the child has a significant infection or a major metabolic abnormality such as

**Liver dysfunction,
Electrolyte imbalance,
Cell membrane damage , or
Damaged biochemical
pathways.**

What are the medical complication in SAM?

- Poor appetite
- Intractable vomiting
- Convulsion
- Lethargy and change in mentation
- Unconsciousness
- High fever $>39^{\circ}\text{C}$
- LRTI
- Dehydration
- Persistent diarrhea
- Severe anemia
- Hypoglycemia
- Hypothermia $<35^{\circ}\text{C}$
- Severe skin lesions
- Eye signs (vit A def)

In patient Management

- Inpatient criteria- failed appetite test or any medical complication
- Grade 3 edema
- Muac<115mm and WFH<-3 +any grade edema

Stabilization phase

- ID life threatening conditions and tx
- Correct specific deficiencies and reverse metabolic derangements
- Feeding with f-75

Managing complications

1. Treating hypoglycemia

-if $bg < 3\text{mmol}(54\text{mg/dl})$

- Prevent with feeding f-75 asap
- Treatment is by 50ml of 10% glucose, feed f-75 Q30 minutes for the first 2 hours then Q2 hours, give Antibiotics
- If the patient is unconscious or lethargic then give IV sterile 10%glucose solution 5mg/kg

- Tx and prevent hypothermia
- How?
- Feed f75 asap
- Keep patient covered
- Warm hands before touching patient
- Give antibiotics
- Monitor t' Q30 minutes in the 1st 2 hours then Q2 hours
- Also check hypoglycemia

- Tx and prevent dehydration

Diagnosed when the patients has dry mouth, sunken eyes, slow skin pinch, irritability, lethargy, **recent diarrhea, vomiting, change in appearance**

**If they have profuse watery diarrhea #cholera
use ORS instead of resomal**

- Prevention
- Feed F75 and replace each loose stool with resomal in pts with no pitting edema
- <2year-50-100ml if >2years-100-200ml
- If they have pitting edema then its 30ml/loose stool
- Encourage BFeeding

- Treating dehydration
- If conscious- 5ml/kg for first 2 hours, if still dehydrated then give 5-10ml/kg/hour alternating with f 75 for 10hours max
- If refusal- NGT
- Preferablly given via **PO**
- The child's RR and PR should slow and there should be voiding with increasing alertness

- Resomal has less Na 45 vs 90
 - More glucose
 - Less potassium 20 vs 40
 - Signs of overhydration
- PR increases by 25, RR by 5 and signs of edema increase, neck vein increased distension

Misdiagnosing and mistreating DHN is the commonest cause of death in SAM undertx

How to make resomal from ORS

- 2L water
- 1 pack ors
- 50gm sugar
- Mineral mix 40 ml/1 scoop

- Treat and prevent infection
- If no complications- Amoxacillin 25mg/kg Q12 hours for 5 days
- If complications- Ampicillin 50mg/kg + Gentamycin 5mg/kg for 7 days followed by amoxacillin for 5 days

Four key steps to manage shock

- I. Give **oxygen**
- II. Give **10% Glucose 5 ml/kg by IV**
- III. Keep the child **warm**
- IV. Give **IV fluids** as follows:-
 1. **15ml/kg over 1hr**
 - RL with 5% glucose , or
 - 0.9% NS with 5% glucose → (125ml of D40% in 1L of NS)
 - Add 20mmol/L KCl
 2. **Observe RR,PR, liver size**
 - ✓ *Slowering of PR & RR* → if increment seen , consider septic shock
 3. **Repeat same dose IV fluid**
 - ✓ 15ml/kg for another 1hr.
 4. **After 2hrs switch to oral /NG tube ReSoMal**
 - ✓ continue with 5-10ml/kg/hr up to 10hrs.

Dx of septic shock

1. A fast weak pulse, with
2. Cold peripheries.
3. Disturbed consciousness
4. Absence of signs of heart failure, AND
5. Failure to improve after the first hour of IV fluids for management of shock in dehydration

Management of septic shock

After 2 bolus IV fluid, if no change

1. Give **10ml/kg whole blood** over 3 hrs.
 - If in HF give packed RBC
 - While waiting for blood transfusion
 - » Give **maintenance IV fluids (4 ml/kg/hour)**
 - Once pulse is strong , & regain consciousness → **stop IV give F-75**
2. Add **second line** Abs
 - If no improvement, Consider **3rd line Abs**, **antifungals** & **anti- staph.**
3. Avoid **physical disturbance** with
 - excess examination & investigations.
 - Neither **wash** nor **transport**.
4. **Sign of liver failure** (jaundice ,purpura) →
 - » Give **vit.K 1mg IM stat.**
5. Increased Abdominal distension give feeding slowly→ if no change
 - » Give **2ml of 50% MgSO4 IM**

Management of anemia

- ❑ Mild or moderate anemia should be treated later,

- ❑ Moderate and severe anemia:- symptoms may appear *between Day two and Day 14 of treatment of malnutrition,*
 - ✓ Temporary excess of fluids will produce dilutional anemia (i.e. **Pseudo-anemia**) → *never to be treated with transfusions*

Management of HF failure

- Stop all oral and IV fluid until HF improves
- Give small amount of sugar + water to see HF improvement
- Give furosemide single dose
- If the cause is severe anemia tx it

Mx of corneal clouding

- For Ulcerations- give vitamin A and atropine
- For clouding give Vit A only
- Vit A <6month- 50,000 6-12 100,000 IU and >12 month give 200,000 IU
- Ulceration + clouding + pus = TTC and bandaging
- If there are signs of vit A def- give high dose Vit A on day 1,2 and 15
- Also give measles vaccine at admission and at discharge

- Dermatitis
 - Unless very sick bathe daily in 1% potassium permanganate and dress oozing areas
 - Protect skin from further damage

PHASE 1

- Feeding during stabilization phase
- Small and frequent F 75 Q2 hours orally or by NGT
- 130ml/kg is daily feed night feeds are important and offer breast milk on demand
- When to use NGT

<80%feed left in 2 consecutive feeds

Has high RR and difficulty swallowing

Painful oral lesion

Cleft palate

Very weak child

Monitoring during stabilization

Monitoring During Stabilization phase

- measure wt & plot on graph / every day
- Assess edema every day
0, +, ++, +++
- MUAC every week
- Vitals Q4h
- Assess & record any standard clinical signs (stools, vomiting, DHN ----) daily

→ During stabilize diarrhea should gradually diminish & children w/ edema should start losing wt

! If Diarrhea continues despite cautious re-feeding — Re-evaluate or worsens mucosal damage & diarrhea are common causes

- ↳ do stool dip or microscopy
- ↳ give Metronidazole
7.5mg/kg Q8hr 7days
even only w/ clinical suspicion

Transition

when to move from stabilize → transition?

- ◇ returned appetite (easily finishes all F-75)
- ◇ subsiding edema (+++ → ++ or ↓)
- ◇ no complications or IV

F-100 / RUT

100x1
130 kcal

How to move from stabilization to transition phase

! IF Diarrhea continues despite cautious re-feeding — Re-evaluate mucosal damage & diarrhea are common causes
 ↳ do stool dip or microscopy
 ↳ give Metronidazole (7.5mg/kg Q8hr 7days) (even only w clinical suspicion)

Transition

when to move from stabilize → transition?

- ♦ returned appetite (easily finishes all F-75)
- ♦ subsiding edema (+++ → ++ or ↓)
- ♦ No serious medical complications or IV

recommended energy intake 100-135 kcal/kg/day

* Introduce **RUTF** gradually alongside F-75, continue offering RUTF

- ✓ The amount is the same as given as OTP.
- ✓ when the child is taking >75% of the daily prescribed amount of RUTF refer to OTP

Feeding procedure → No RUTF to infants 0-6 mo

- Feed **Q4h**
- Breast milk before RUTF & on demand
- offer clean & safe H₂O during & after RUTF
- provide the same amount of F-75 as in stabilize (Q3-4hr)

when the pt finishes 50% of RUTF, reduce F-75 by 50%
 & stop F-75 when the child finishes 75-100% of daily RUTF.

→ Cerebral Palsy

100ml F100 = 200 RUTF

130ml/kg F-100

RUTF

100ml F100 = 200 RUTF

100 > 120 kcal/day

F-100 / RUTF

dangerous

physical

know

Sg proteas

who can't tolerate RUTF

difficulty swallowing

When to return to stabilization, move to rehabilitation

can RUTF stay at SC,
Other showing progress → OTP

When to Return to Stabilization Phase

- weight gain of $>10\text{g/kg/day}$ w respiratory rate $\uparrow\uparrow$ (fluid excess)
- Increasing or developing edema ✓
- Rapid $\uparrow$ in Liver size ✓
- Tense abd. distension ✓
- Any sign of fluid overload ✓
- complications necessitating IV infusion
- significant refeeding diarrhea → weight loss
- The need for NGT

Handwritten calculations:
 $100 = 700$
 $150 = ?$
 $1320 \div 100 = 13.2$
 $1320 \div 100 = 13.2$
 $1320 \div 100 = 13.2$
 $1320 \div 100 = 13.2$

When to Move to Rehabilitation — OTP

- good appetite, take $>75\%$ of daily RUTF
- edema is reduced to moderate or if wasting w edema, edema should completely disappear
- ~~improving~~ resolving complications
- Clinically well & Alert

Rehabilitation Phase

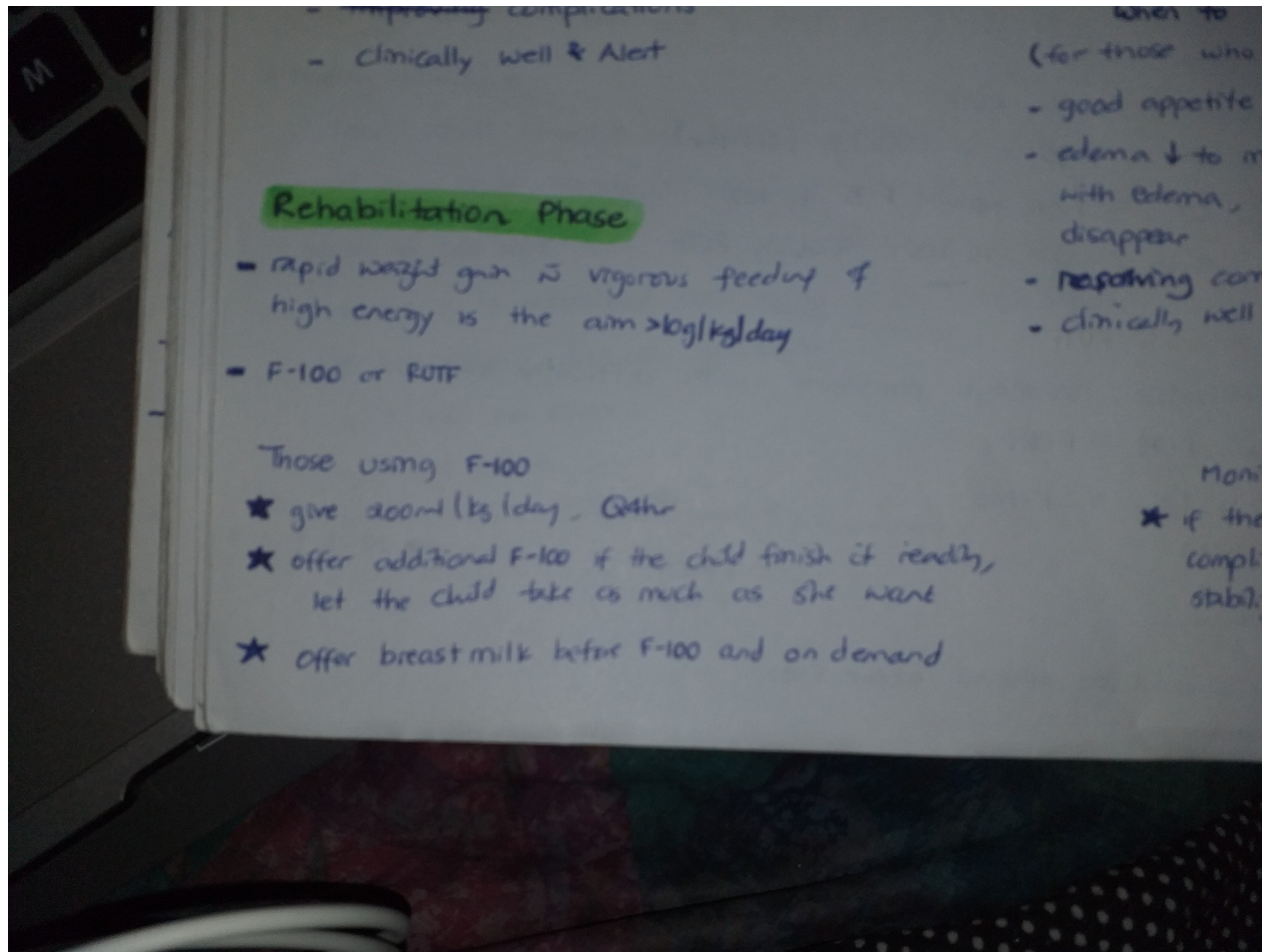
- rapid weight gain w vigorous feeding of high energy is the aim $>10\text{g/kg/day}$

When to move to Rehabilito — SC

(for those who can't tolerate RUTF)

- good appetite, take all F-100 prescribed
- edema $\downarrow$ to moderate or if wasting with edema, edema should completely disappear
- resolving complications
- clinically well & alert

The rehabilitation phase



Deworming Iron Danger signs

Deworming Agent

- give at the start of phase 2
only for kids who can ambulate (≥ 2 yrs)

give single dose if she didn't get
within the previous 6 months

Mebendazole

Albendazole

500mg
tablet,

400mg
tablet,

Age ≥ 5 25ml syrup = 5 tsp
(100/5ml)

20ml syrup = 4 tsp
(100ml/5ml)

Iron

- If the child is on F-100
add iron in phase 2

* RUTF has enough iron

Danger Signs

- anorexia
- Δ in mentation
- Jaundice
- cyanosis
- difficulty breathing
- Drowsiness
- new edema
- abd. distension
- large weight Δ
- $\uparrow$ Vomiting
- petechiae

→ Add 1 crushed tablet of ferrous Sulphate (300mg)
to each 2ltrs - 2.4ltrs of F-100

* For 1-1.2ltrs of F-100, dilute 1 tab FeSO_4
in 4ml H_2O & add 2ml of the solution &
for 0.5-0.6ltrs of F-100, add 1ml.

Exceptions are already addressed

* VitA, folic acid, Zn, Cu are present in all the
therapeutic food; so no additional dose is needed
* $\parallel$ Double-Dosing

Providing Sensory stimulation & Emotional Support

- throughout the period of SC $\rightarrow$ bright ^{colored room} + decoration
 $\rightarrow$ tender loving care, cheerful env't, structured play therapy for 15-30 min/day
 $\rightarrow$ maternal involvement

FAILURE TO RESPOND TO TREATMENT

The first dose of measles often
doesn't give protective response.

- It is given to ameliorate the
severity of incubating measles
& partially protect from nosocomial
measles

(since the SAM pt's response is
very weak, specially cell mediate)

* Unnecessary in OTP

- The 4th wk dose is given
to provoke protective Ab

Failure to respond to tx

FAILURE TO RESPOND TO TREATMENT

criteria

Failure to regain appetite by Day 4

Failure to start to lose edema <<

Edema still present Day 10

Failure to gain at least 5g/kg Day 10

What to do for tx failure

More Frequent causes of FTRTT

- Health facility
 - lack of adherence to Rx protocols
 - poor env't
 - insufficient / inadequately trained staff
 - inadequate supervision
 - inaccurate wt scale
 - incorrect feed preparation or given
 - Child's
 - Infections
 - amebiasis, giardiasis, dysentery
 - pneumonia, TB, UTI, O. media, malaria
 - HIV, schistosomiasis, hepatitis
 - underlying conditions
 - Down syndrome, cerebral palsy,
 - inborn metabolic errors.
- c
- insufficient feeds
 - psychological trauma — refugee, HIV
 - Vitamin & mineral deficiencies
 - Malabsorption
 - Rumination

Actions required to be taken

- ★ keep accurate records Age, wt, MUAC, Lt/Lt ----
- ★ systematically examine common causes of FTRTT
- ★ if the above maneuvers couldn't pick the problem recalibrate scales, train staffs ---

- Discharge SC → OTP procedure

W ★ Give feedback on feed preparation

Dis

Failure to thrive

II) Failure to thrive

Tx: make mealtime positive experience
• don't force feed

- Identify underlying cause
- Avoid mealtime disruption
- Formula amount + water
- schedule meals
- offer solid before liquid
- ↑ protein
- psychological
- Antacid - GERD
- cardiac / lung - Tx.

C/F

- History
 - Maternal
 - NICU
 - Infant - feeding & devt
 - Food / Nutritional hx
 - SES
 - caregiver symptoms
 - Abuse, neglect
 - family educat.
- Illness - Immunization
 - Wt loss - Serostat
 - SUBTI
 - LETI
 - N.V
 - Diarrhea
 - CNS abnormal.
 - flaccid arms
 - abnormal gait

Cause

Inadeq. Intake

- poverty
- Mixing formula
- cleft lip
- CPalsy
- GERD
- HPS stenosis

Malabsorption

- Lactose intolerance
- C. Fibrosis
- Celiac disease
- NEC

Increased demand

- Hyperthyroidism
- TB
- HIV
- TORCH
- CHD
- DM
- Chronic Lung Dis.

PIE - normal vital signs

Intact / mildly delayed milestone

Anthropometry

Weight Length
WFH H/C BMI

General app - activity, appearance

Skin & hair - texture
nails
hygiene
trauma

NECK CLAP

- Thyroid
- LAD

Chest

- CRS

Abd

- Distension
- Organomeg.
- bowel sound

MSS

- Edema
- muscles

- Gait
- Proximal force
- Tone
- Specificity

HEENT

- Frontal bossing
- Dysmorphic feat.
- pallor
- Dental caries
- Glossitis
- thrush

Situation where a child is undernourished
Not meeting standards of growth

Vitamin D deficiency

Condition	Genetics	Ca	Phos.	Alk Phos	PTH	Vit D	1,25 (OH)VitD
Vitamin D Resistant Rickets (Hypophosphatemic)	X linked dominant	–	↓	↑	–	–	
Vitamin D Deficiency Rickets (Nutritional)	Nutritional	↓	↓	↑	↑	↓	
Type I Vitamin D Dependent	Auto. Recessive	↓	↓	↑	↑		↓ ↓
Type II Vitamin D Dependent	Auto. Recessive	↓	↓	↑			↑ ↑
Hypophosphatasia	Auto. Recessive	↑	↑	↓ ↓	–	–	
Renal Osteodystrophy	Renal Disease	↓	↑	↑	↑		
Hyperparathyroidism	90% adenoma	↑	↓	↑	↑		

Types of rickets

Calcipenic rickets

- **Vitamin D deficiency or resistance**
 - Dietary deficiency
 - Malabsorption
 - Lack of sunlight exposure
 - Defect in 25 hydroxylation of vitamin D (e.g., liver disease, medications such as phenytoin)
 - Failure of 1 hydroxylation of vitamin D due to inherent deficiency of 1 alpha hydroxylase secondary to defects in the 1 alpha hydroxylase gene (VDDR I)
 - End-organ resistance to vitamin D (VDDR II)
- **Calcium deficiency**
- **Renal rickets secondary to CKD**

Phosphopenic rickets

- Renal tubular phosphate loss**
- Isolated phosphate loss secondary to genetic mutations:
 - XLHR
 - ARHR
 - ADHR
 - Hypophosphatemic rickets with hypercalciuria
- Renal Fanconi syndrome
- Dietary phosphate deficiency
- Phosphate malabsorption

In the case of nutritional and type 1 vitamin D dep

Laboratory findings

Elevated:

Alkaline phosphatase
Parathyroid hormone
Dihydroxyvitamin D

Decreased:

Calcium
Phosphorus
Hydroxyvitamin D

R/F

- Maternal vitamin D def
- Prematurity
- Prolonged BF without supplementation
- Avoidance of sun exposure
- Dark skin
- Age 3 months- 3 years
- Drugs (anticonvulsants, ketoconazole, HIV drugs)
- Malabsorption, bowel surgery and gastrectomy

C/F

EARLY

- Craniotabes
- Rachitic rosary
- Wrist and ankle widening
- Double malleoli

LATE

- Harrisons groove
- Atelectasis leading to pneumonia
- Bow legs
- Greenstick#
- Frontal bossing
- Pigeon chest
- Knocked knee
- Hyptonia
- Caput quadratum
- Delayed fontanel closure and tooth eruption

X ray features

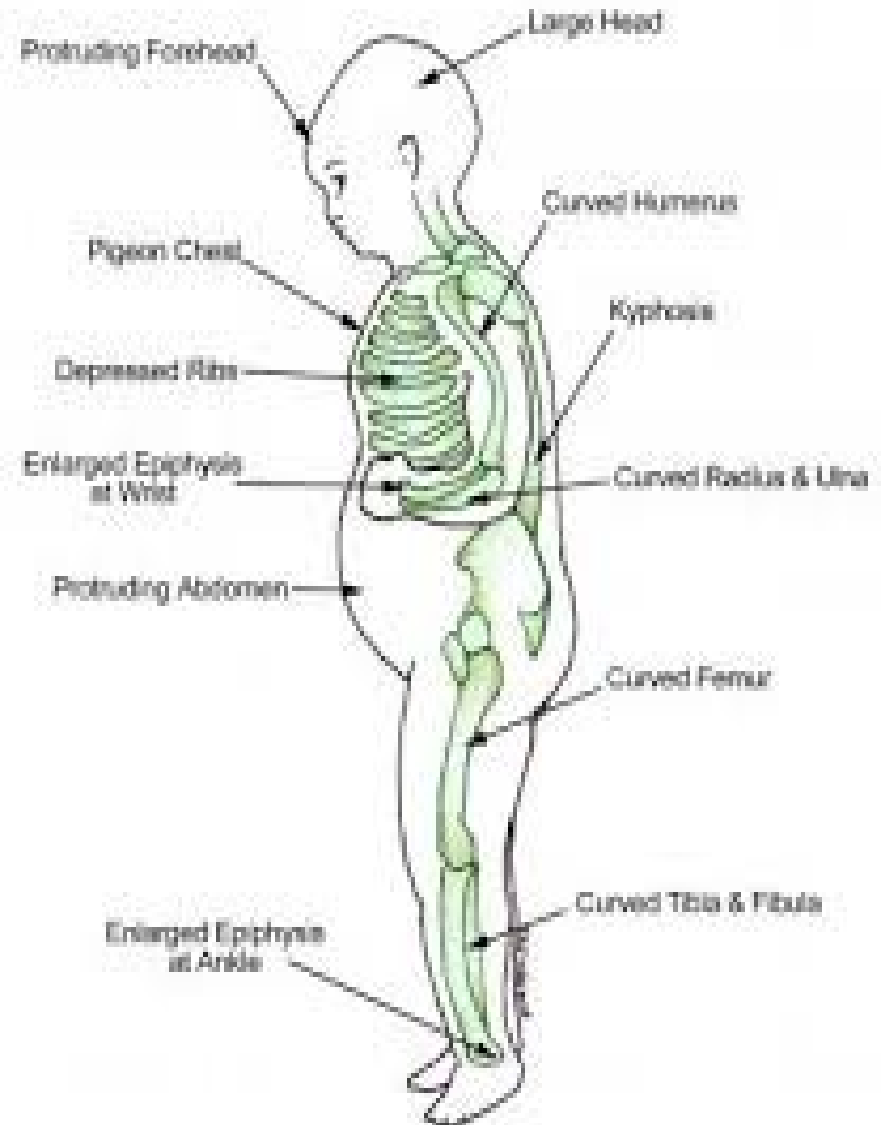
- Fraying-the edge of the metaphysis loses its sharp borders
- Cupping- metaphysis lose its normal convex or flat structure
- Growth plate thickens and the distal metaphysis widens
- Coarse diaphyseal trabeculation and generalized rarefaction

Clinical manifestation



Fraying & cupping







Rachitic rosary



- Harrison groove.



- A picture is shown with a child who have rachitic rosary and harrisons sulcus
 - a. What do u see
 - b. What is ur likely diagnosis
 - c. What lx do u needHow do u prevent

Investigation

- Serum ca and P
- ALP
- PTH
- 25 D-OH Vitamin D (hydroxycholecalciferol)
- 1,25-D OH vitamin D
- Urine analysis
- Creatinine and BUN
- LFT
- Wrist Xray
- Based on presence of complications- chest x ray, ECHO

Management

- Depends on the cause
- Non pharmacologic management is optimum sun exposure
- Pharmacologic- Mega dose of Vitamin D3 600,000 IU IM healing proven by wrist X ray after 2-4 weeks

- Natural and artificial light of appropriate wavelength
- Oral administration of vitamin D
 - 50-150 microgram Vit.D3 daily
[cholecalciferol]
 - 0.5-2 microgram 1,25 dihydrocholi.
[calcitriol]
- A single mega dose of vit D-600,000IU IM
- Healing is rapid within 2-4 weeks

Complications

- Pneumonia
- Hypocalcemia tetany
- Pathologic #
- Atelectasis
- Pulmonary HTN
- Cor pulmonale

Prevention

- Exposure to ultraviolet light
- Vitamin D supplementation
- Vitamin D should also be administered to pregnant and lactating mothers

Vitamin A def

BITOT'S SPOT

Triangular pearly white or yellowish foamy spots on bulbar conjunctiva on either side of the cornea.



17

Bitot's spots – foamy, bubbly, or cheesy accumulation



8)

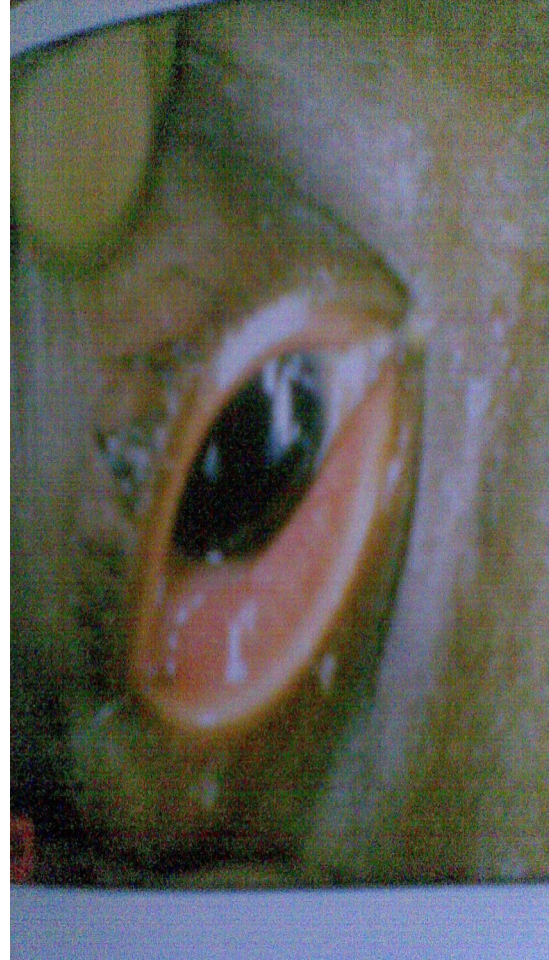
1. What do u see

2. What is ur likely diagnosis

3. How do u prevent



- Corneal scar



Cause

- Inadequate consumption
- Problems of absorption
- Problem in conversion/utilization of vitamin A
- Repeated infections or diseases such as measles or diarrhea
- Absence of food containing oil or fat in the diet

What does Vit A do?

- Photo transduction
- Prevention of xerophthalmia
- Growth
- Bone
- Growth
- Reproduction- for normal pregnancy and fetal dev't
- Cellular differentiation

Sources

❖ <u>Source (per 100gm)</u>	<u>vit.A (mic.gm)</u>
■ Fish liver oil	145,000
■ Liver, ox	840
■ Butter, cow's	730
■ Kidney, ox	300
■ Egg, hen	200
■ Chicken	85
■ Milk, whole cow's	40
■ Milk, goat	25

<u>Source (per 100gm)</u>	<u>vit.A(mic.gm)</u>
■ Carrot, raw	1100
■ Mango	400
■ Lettuce	325
■ Sweet potato, raw	510
■ Avocado, raw	90
■ Tomatoes, raw	75
■ Papaya, raw	75

CLINICAL FEATURES

- WHO CLASSIFICATION :
 - XN NIGHT BLINDNESS
 - X1A CONJUCTIVAL XEROSIS
 - X1B BITOT'S SPOTS
 - X2 CORNEAL XEROSIS
 - X3A COR. ULCER $< 1/3$ OF COR. SURFACE
 - X3B COR. ULCER $> 1/3$ OF COR. SURFACE
 - XS COR. SCAR
 - XF XEROPHTHALMIC FUNDUS

IX and DX

- Clinical
- Laboratory test
 - plasma vitamin A level – low
(normal 20 -50 microgram/100ml)

Therapy

- Vitamin A capsule on two consecutive days immediately and another dose within 1-4 weeks

Dose -200,000IU above one year

-100,000IU 6 – 12 month

-50,000IU less than 6 months

Prevention

- Increase consumption of dark green leafy vegetables. Egg, livers, fat of fish and meat and cod liver oil can be provided
- Vitamin A should be supplemented in malnutrition, diarrhea, measles and acute respiratory infection.
- Distribution of vitamin A capsule should be given to the community.
 - One capsule every 6 months up to 6 year of age
- One drop of vitamin A (25,000IU) for every child with immunization schedule